

LAB EXPERIMENTS

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Experiment – 18 :

```
import pandas as pd
```

```
data = {  
    "school": ["S1", "S2", "S3", "S4", "S5", "S6"],  
    "class": ["s001", "s002", "s003", "s001", "s002", "s004"],  
    "name": [  
        "Alberto Franco",  
        "Gino Mcneill",  
        "Ryan Parkes",  
        "Eesha Hinton",  
        "Gino Mcneill",  
        "David Parkes"  
    ],  
    "age": [12, 12, 13, 13, 14, 14],  
    "height": [173, 192, 186, 167, 151, 159],  
    "weight": [35, 32, 33, 30, 31, 32]  
}
```

```
df = pd.DataFrame(data)
```

```
print("Original DataFrame:")
```

```
print(df)
```

```
# Split into groups based on school and class
```

```
groups = df.groupby(["school", "class"])
```

```
# Display each group
```

```
for name, group in groups:
```

```
    print("\nGroup:", name)
```

```
    print(group)
```

```
Original DataFrame:
  school class      name  age  height  weight
0     S1  s001  Alberto Franco   12    173     35
1     S2  s002   Gino Mcneill   12    192     32
2     S3  s003   Ryan Parkes   13    186     33
3     S4  s001   Eesha Hinton   13    167     30
4     S5  s002   Gino Mcneill   14    151     31
5     S6  s004   David Parkes   14    159     32

Group: ('S1', 's001')
  school class      name  age  height  weight
0     S1  s001  Alberto Franco   12    173     35

Group: ('S2', 's002')
  school class      name  age  height  weight
1     S2  s002   Gino Mcneill   12    192     32

Group: ('S3', 's003')
  school class      name  age  height  weight
2     S3  s003   Ryan Parkes   13    186     33

Group: ('S4', 's001')
  school class      name  age  height  weight
3     S4  s001   Eesha Hinton   13    167     30

Group: ('S5', 's002')
  school class      name  age  height  weight
4     S5  s002   Gino Mcneill   14    151     31

Group: ('S6', 's004')
  school class      name  age  height  weight
5     S6  s004   David Parkes   14    159     32
PS C:\Users\Vedvyass M\Music\Nutri-vision>
```

Experiment – 19 :

```
import pandas as pd
```

```
# World Alcohol Consumption Dataset
```

```
data = {  
    "Year": [1986, 1986, 1985, 1986, 1987],  
    "WHO region": [  
        "Western Pacific",  
        "Americas",  
        "Africa",  
        "Americas",  
        "Americas"  
    ],  
    "Country": [  
        "Viet Nam",  
        "Uruguay",  
        "Cte d'Ivoire",  
        "Colombia",  
        "Saint Kitts and Nevis"  
    ],  
    "Beverage Types": [  
        "Wine",  
        "Other",  
        "Wine",  
        "Beer",  
    ]  
}
```

```
        "Beer"  
    ],  
    "Display Value": [  
        0.00,  
        0.50,  
        1.62,  
        4.27,  
        1.98  
    ]  
}
```

```
# Create DataFrame
```

```
df = pd.DataFrame(data)
```

```
# Display the dataset
```

```
print("World Alcohol Consumption Dataset:")
```

```
print(df)
```

```
# Display dimensions / shape
```

```
print("\nShape of the dataset:")
```

```
print(df.shape)
```

```
# Extract column names
```

```
print("\nColumn names:")
```

```
print(df.columns.tolist())
```

```

● PS C:\Users\Vedvyass M\Music\Nutri-vision> c::; cd 'c:\Users\Vedvyass M\Music\Nutri-vision\vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bundled\libs\debugpy\
World Alcohol Consumption Dataset:
  Year  WHO region  Country Beverage Types  Display Value
0  1986  Western Pacific  Viet Nam  Wine  0.00
1  1986  Americas  Uruguay  Other  0.50
2  1985  Africa  Cte d'Ivoire  Wine  1.62
3  1986  Americas  Colombia  Beer  4.27
4  1987  Americas  Saint Kitts and Nevis  Beer  1.98

Shape of the dataset:
(5, 5)

Column names:
['Year', 'WHO region', 'Country', 'Beverage Types', 'Display Value']
○ PS C:\Users\Vedvyass M\Music\Nutri-vision>

```

Experiment – 20 :

```
import pandas as pd
```

```
# Create DataFrame
```

```
data = {
    "Country": [
        "Viet Nam",
        "Uruguay",
        "Cte d'Ivoire",
        "Colombia",
        "Saint Kitts and Nevis"
    ],
    "Beverage": [
        "Wine",

```

```

        "Other",
        "Wine",
        "Beer",
        "Beer"
    ]
}

df = pd.DataFrame(data)

# Given substring
substring = "Col"

# Find the index where the substring occurs
index = df["Country"].str.find(substring)

print("DataFrame:")
print(df)

print("\nIndex of substring:", substring)
print(index)

```

```

PS C:\Users\Vedvyass M\Music\Nutri-vision> c::; cd 'c:\Users\Vedvy
ss M\.vscode\extensions\ms-python.debugpy-2026.6.0-win32-x64\bund
DataFrame:
   Country Beverage
0   Viet Nam   Wine
1   Uruguay   Other
2  Cte d'Ivoire   Wine
3   Colombia   Beer
4 Saint Kitts and Nevis   Beer

Index of substring: Col
0   -1
1   -1
2   -1
3    0
4   -1
Name: Country, dtype: int64
PS C:\Users\Vedvyass M\Music\Nutri-vision>

```